

## Supplementary Fig. 1 | Public-data identification and eligibility flow

Records identified through database searches\nGEO n = 42; PubMed n = 3



Additional record from citation/accession checking\nn = 1



Records screened\nExcluded at title/summary\nn = 34



Excluded after full assessment\nn = 5\nnone donor (1); split series (2);\nglass only (1); strain/vibration (1)



Studies included in quantitative synthesis\nn = 7



Independent validation source\nGSE147390; one donor; n = 5,329 osteoblasts